Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial\mathcal{V}}{\partial\varphi}=\tfrac{2}{n}\frac{1}{\mathrm{Tr}(\mathcal{M}^2)}\,\mathcal{M}_{mn}\frac{\partial\mathcal{V}}{\partial\mathcal{M}_{mn}}\,,\quad\Rightarrow\mathcal{M}^{mn}\mathcal{M}_{pq}\frac{\partial\mathcal{V}}{\partial\mathcal{M}_{pq}}-\mathrm{Tr}(\mathcal{M}^2)\frac{\partial\mathcal{V}}{\partial\mathcal{M}_{mn}}=0\,.$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\int DV\exp(\frac{1}{2}\int d^4x\varepsilon^{\mu\nu\tau\lambda+}W^{\alpha\beta+}_{\mu\nu}G^{\sigma\rho}_{\tau\lambda}\epsilon_{\alpha\beta\sigma\rho}-\frac{1}{2}\int d^4x\varepsilon^{\mu\nu\tau\lambda-}W^{\alpha\beta-}_{\mu\nu}G^{\sigma\rho}_{\tau\lambda}\epsilon_{\alpha\beta\sigma\rho}).$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$I=\int_\Sigma d^3x\sqrt{\gamma}\left(R(\gamma)-\frac{1}{2}\Big((DU)^2-e^{-2U}(DV)^2\Big)-\frac{1}{2}\Big((D\Phi)^2-e^{-2\Phi}(D\Psi)^2\Big)-(DT)^2+e^{-(U+\Phi)}V(T)\right)$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$2\frac{\int_{b=\theta_2}^{b=\theta_1}e^{\frac{\rho rs\cos(b)}{1-\rho^2}}\,\underset{\cdot}{\mathrm{b}}}{2\pi\int_0^{2\pi}e^{\frac{\rho rs\cos a}{1-\rho^2}}\,\underset{\cdot}{\mathrm{a}}}\leq 2\frac{\int_{b=0}^{b=\theta_1-\theta_2}e^{\frac{\rho rs\cos(b)}{1-\rho^2}}\,\underset{\cdot}{\mathrm{b}}}{2\pi\int_0^{2\pi}e^{\frac{\rho rs\cos a}{1-\rho^2}}\,\underset{\cdot}{\mathrm{a}}}<\frac{\theta_1-\theta_2}{2\pi^2}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\frac{d}{d\lambda}S_{\omega,\mathbf{c}}(\Phi_0^\lambda)=-(\sqrt{\lambda}-1)(2L(\Phi_0)+\mathbf{c}\cdot\mathbf{P}(\Phi_0))\left(\lambda+\frac{\mathbf{c}\cdot\mathbf{P}(\Phi_0)}{2L(\Phi_0)+\mathbf{c}\cdot\mathbf{P}(\Phi_0)}\sqrt{\lambda}+\frac{\mathbf{c}\cdot\mathbf{P}(\Phi_0)}{2L(\Phi_0)+\mathbf{c}\cdot\mathbf{P}(\Phi_0)}\right)$$